WIZFORGE AI - CODE REVIEW REPORT

payroll/Employee.java

Java | 21 Oct 2026, 07:24 UTC

Summary

Quality score	**40/100 (grade F)**
Score after fixes	**80/100 (grade B)**
Lines of code	**36**
Functions	**3**
Average complexity	**1.7**
Findings	**2 critical, 1 high, 1 info**

Findings (4)

CRITICAL **Missing semicolon (line 17, CF001)**
This statement does not end with ';', so the file will not compile.
Fix: Add ';' at the end of the statement.

CRITICAL **Null dereference of 'e' (line 47, CF003)**
'e' is set to null on line 46 and used here without being assigned or checked, which crashes at runtime (NullPointerException, segmentation fault or TypeError).
Fix: Give 'e' a real value first, or guard the access with a null check.

HIGH **Possible typo: 'totalbonus' (line 44, CF002)**
'totalbonus' differs from 'totalBonus' only by letter case. Most of these languages are case-sensitive, so this is likely an undefined name.
Fix: Did you mean 'totalBonus'?

INFO **Console printing (6×) (line 38, JV002)**
System.out/err printing in application code is hard to control. Also on lines 39, 40, 41, 44, 47.
Fix: Use a logging framework (SLF4J).

Automatic fixes

- line 17: Added the missing semicolon
- line 44: Renamed 'totalbonus' to 'totalBonus'

Source code

```
 1 |
 2 | import java.util.*;
 3 |
 4 | class Employee {
 5 |     String name;
 6 |     double salary;
 7 |
 8 |     Employee(String name, double salary) {
 9 |         this.name = name;
10 |         this.salary = salary;
11 |     }
12 |
13 |     double calculateBonus() {
14 |         if (salary >= 50000) {
15 |             return salary * 0.20;
16 |         } else {
17 |             return salary * 0.10
18 |         }
19 |     }
20 | }
21 |
22 | public class Main {
23 |     public static void main(String[] args) {
24 |
25 |         ArrayList<Employee> employees = new ArrayList<>();
26 |
27 |         employees.add(new Employee("Arun", 60000));
28 |         employees.add(new Employee("Meera", 45000));
29 |         employees.add(new Employee("Rahul", 75000));
30 |
31 |         double totalBonus = 0;
32 |
33 |         for (Employee emp : employees) {
34 |             double bonus = emp.calculateBonus();
35 |
36 |             totalBonus += bonus;
37 |
38 |             System.out.println("Employee: " + emp.name);
39 |             System.out.println("Salary: " + emp.salary);
40 |             System.out.println("Bonus: " + bonus);
41 |             System.out.println("-------------------");
42 |         }
43 |
44 |         System.out.println("Total Bonus: " + totalbonus);
45 |
46 |         Employee e = null;
47 |         System.out.println(e.name);
48 |     }
49 | }
```